Project Development Phase
Performance Test

Date	01 March 2026
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Project Name	Heart Disease Analysis
Maximum Marks	

Model Performance Testing:

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	**Column Description:** • **HeartDisease** – Target variable (Yes/No). • **BMI** – Body Mass Index value. • **Smoking** – Whether the person smokes. • **AlcoholDrinking** – Whether the person drinks alcohol. • **Stroke** – Whether the person has had a stroke. • **PhysicalHealth** – Days of poor physical health in the last month. • **MentalHealth** – Days of poor mental health in the last month. • **DiffWalking** – Difficulty walking or climbing stairs. • **Sex** – Gender of the person. • **AgeCategory** – Age group of the person. • **Race** – Race category. • **Diabetic** – Whether the person is diabetic. • **PhysicalActivity** – Physical activity in the last 30 days. • **GenHealth** – General health rating. • **SleepTime** – Average hours of sleep. • **Asthma** – Whether the person has asthma. • **KidneyDisease** – Whether the person has kidney disease. • **SkinCancer** – Whether the person has skin cancer.
2.	Data Preprocessing	• Removed duplicates from the dataset • Checked and handled missing or invalid values • Converted categorical fields (Sex, AgeCategory, Race, etc.) into proper labels • Ensured correct data types for measures and dimensions in Tableau • Standardized column names for clarity • Created derived fields where necessary (e.g., BMI groups, health categories)

3.	Utilization of Filters	• Age Category • Sex • Race • Smoking • Alcohol Drinking • Diabetic • Physical Activity • General Health (GenHealth) • Sleep Time • Asthma • Kidney Disease • Skin Cancer • Stroke • Difficulty Walking • BMI Range • Heart Disease (Target)
4.	Calculation fields Used	• **BMI Category** (Underweight / Normal / Overweight / Obese) • **Heart Disease Indicator** (Converted 0/1 to Yes/No) • **Risk Level** (Based on BMI, Smoking, Alcohol, and Age) • **Sleep Category** (Low / Normal / High Sleep)
5.	Dashboard design	No of Dashboards- 03 • Comprehensive Heart Disease Risk Analysis by Demographics and Lifestyle Factors • Heart Health Analysis • Multi-Factor Heart Disease Analysis
6	Story Design	No of Visualizations – 16 story points • The story includes insights on **Gender**, **Age**, **BMI**, **Diabetes**, **Stroke**, **Smoking**, **Alcohol**, **Physical Activity**, **General Health**, **Sleep**, **Race**, and **Mental Health** • Uses multiple chart types: bar charts, stacked charts, pie chart, lollipop chart, tree maps, jitter plots, and text panels. • Provides a complete progression from • **Introduction → Key Risk Factors → Combined Analysis → Summary & Recommendations**.